

MST-Technologies and Processes

April, 28th, 2020

Duration: 120 min.

Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task

Cheating

- any attempt of cheating results in immediate failure of the exam
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
- please see:
https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams?set_language=en
- you find the relevant forms here:
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>

Necessary Points: **55 of 110 in total**

Points: _____

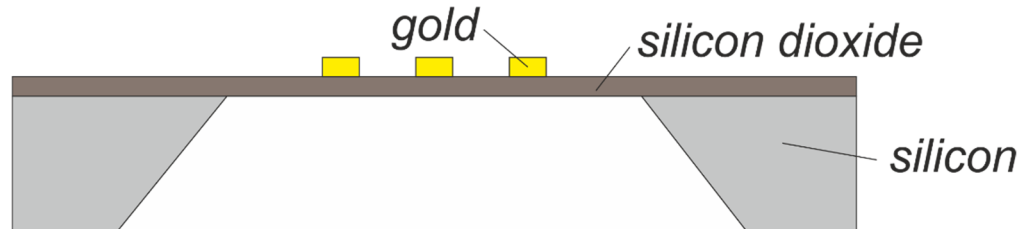
Result: _____



1. Processes

20

The following components needs to be manufactured. It is a membrane-supported microheater. The substrate material is silicon, the membrane is made from silicon (di)oxide, the heater itself from gold.



- | | | |
|-----|--|---|
| 1.1 | Which wafer cut do you chose and why? | 2 |
| 1.2 | Sketch the manufacturing steps required to make the silicon dioxide layer; pay attention to the fact that the layer is only on one side of the substrate | 4 |
| 1.3 | Sketch the process for generating the cut-out on the backside which extends to the oxide layer | 8 |
| 1.4 | Sketch the steps required to make the gold structure on the frontside | 6 |

2. Thermal oxidation of silicon

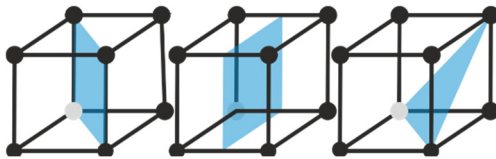
20

- | | | |
|-----|--|---|
| 2.1 | State the reaction for wet chemical oxidation of silicon | 2 |
| 2.2 | State the reaction for dry chemical oxidation of silicon | 2 |
| 2.3 | State the reaction for pyrogenic wet chemical oxidation of silicon | 2 |
| 2.4 | Sketch and explain the setup used for pyrogenic wet chemical etching | 4 |
| 2.5 | Which are the two stages during thermal oxidation of silicon? | 2 |
| | Which is the limiting physical quantity in each of these stage? | 2 |
| | What is the correlation between layer height and time in each of these stages? | 2 |
| 2.6 | Does doping increase or decrease the oxidation rate and why? How is this process called? | 4 |



3. Semiconductor materials 26

- 3.1 Name two common semiconductor materials 2
- 3.2 What is their most outstanding physical properties (for which they would be most likely chosen in an application) 2
- 3.3 What is an amorphous material (sketch and explain)? 2
What is a semicrystalline material (sketch and explain)? 2
What is a polycrystalline material (sketch and explain)? 2
What is a crystalline material (sketch and explain)? 2
- 3.4 Indicate the following crystal planes by their Miller indices 6



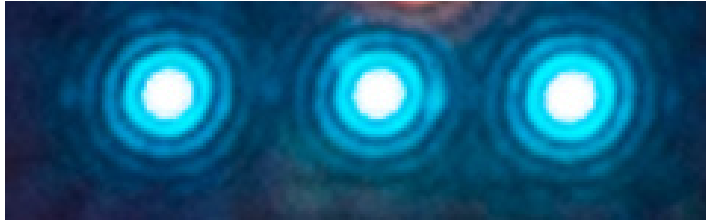
- 3.5 Sketch and explain two typical two-dimensional effect in crystalline materials 4
- 3.6 Sketch and explain two typical three-dimensional effect in crystalline materials 4

4. Photolithography 26

- 4.1 Which two properties of light than can be modulated by the photomask? 2
- 4.2 There are three distinct types of photolithographic setups 6
 - Sketch and explain them 6
 - Give the formular for the minimum achievable feature resolution for each case? 3
- 4.3 Which are the three most commonly used transparent substrates for photomasks? 3

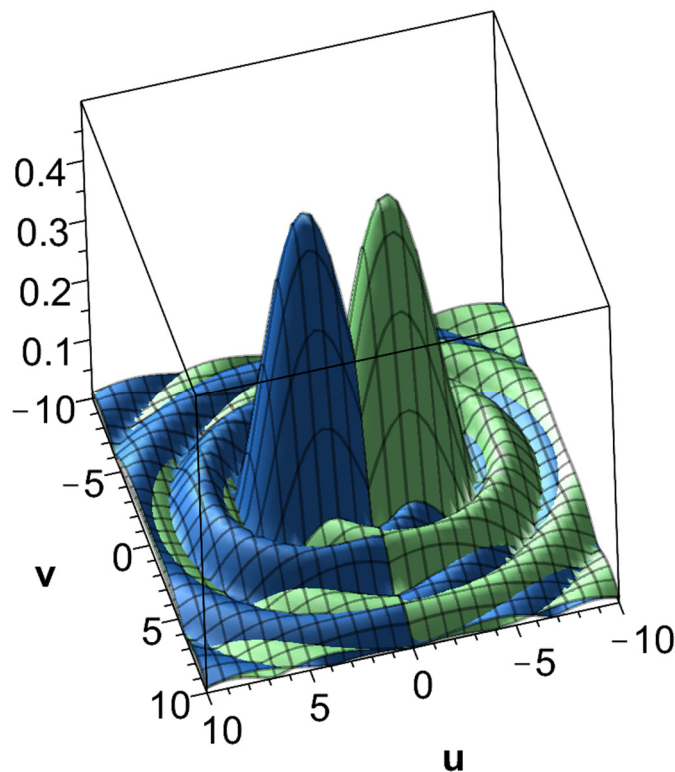


- 4.4 Point sources do not appear as points in the far-field in projection lithography, they form the so-called “Airy disc pattern”:



Are these two point sources resolvable/resolved? Explain your answer

2



When are two features resolvable – state the criterion and its formula which must be fulfilled in order for two features to be resolvable?

4

- 4.5 Name the two relevant classes of photoresists

2

- 4.6 Sketch the workflow for both photoresists as well as the final structures when exposing the respective photoresist with a single-slit mask

4



5. Etching

18

5.1 Sketch and explain the following etching technologies:

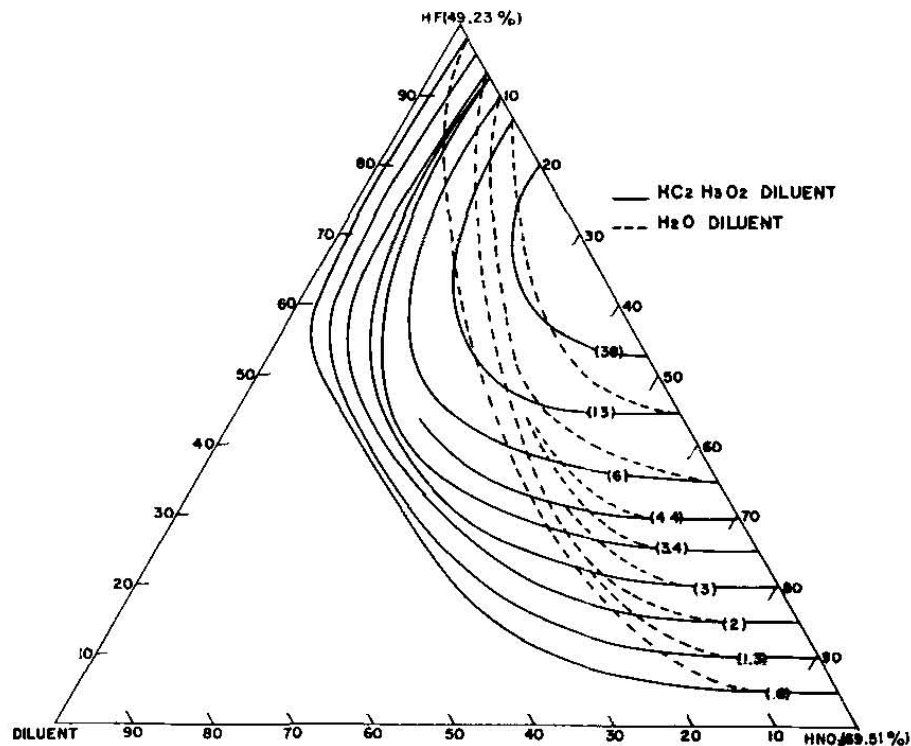
- plasma etching
- sputter etching

5

5

5.2 Given the following etching diagram state three etching solutions which give an etching rate of 3 ($\mu\text{m}/\text{min}$) in acetic acid (solid lines) as diluent

6



5.3 Which one would you prefer and why?

2



Declaration of the absence of a suspected coronavirus infection when participating in an on-site test at the University of Freiburg

We ask you to fill out this form to protect the health of your fellow examination candidates and supervisory staff. If you are unable to sign the form because you are suspected of being infected with the coronavirus, please exercise your right to withdraw from the examination. In accordance with the regulations in the study and examination regulations, the application for permission to withdraw from the examination must be submitted to the responsible office without delay and with appropriate proof.

First and last name

Matriculation number

Course of study

Exam subject

Exam date

Appointed exam time

I hereby confirm to the best of my knowledge that I

1. do not feel any symptoms that are signs of an infection with the coronavirus (these include in particular fever and dry cough, but may also indicate an infection, for example a cold, shortness of breath, sore throat and limbs, tiredness and chills) and for which there are no other explanations
2. have not tested positive for the coronavirus in the past 14 days
3. am not under an officially authorized quarantine
4. have not had contact with a person proven to be infected with the coronavirus within the last 14 days

I am aware that according to § 7 of the regulation of the state government on infection-protective measures against the spread of the virus SARS-Cov-2 (Corona regulation - CoronaVO), a ban on entering the University applies to persons who have had contact with infected persons in the last 14 days, have symptoms of a respiratory tract infection or increased temperature. A breach of the ban on entering the premises constitutes an administrative offense in accordance with § 9 No. 13 CoronaVO, which can be punished by a fine.

Furthermore, I am aware that the violation of an officially ordered quarantine according to §§ 75 paragraph 1 number 1, 30 paragraph 1 of the Infection Protection Act is punishable with imprisonment for up to two years or with a fine.

Date and location

Signature



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